Lab-1
1st January 2026

I have done this experiment in MySQL workbench
Here I have created all the tables in a database named *product_management_db*

CODE:

The code for this given Experiment is as below:

Create the following tables (Don't specify any constraints):

A. Category_details (category_id numeric (2), category_name varchar (30))

B. Sub_category_details (sub_category_id numeric(2), category_id numeric(2),sub_category_name varchar(30)

C. Product_details (Product_id numeric (6), category_id 3 numeric(2), sub_category_id numeric(2), product_name, varchar(30))

```
CREATE DATABASE product_management_db;
USE product_management_db;

CREATE TABLE Category_details(
   category_id NUMERIC(2),
   category_name VARCHAR(30)
   );

CREATE TABLE Sub_category_details(
   sub_category_id NUMERIC(2),
   category_id NUMERIC(2),
   sub_category_name VARCHAR(30)
   );

CREATE TABLE Product_details(
   product_id NUMERIC(6),
   category_id NUMERIC(2),
   sub_category_id NUMERIC(2),
   product_name VARCHAR(30)
   );
```

1. Add a primary key constraint (without any constraint name) on column category_id of category_details table.

```
ALTER TABLE Category_details
ADD PRIMARY KEY (category_id);
```

2. Add a primary key constraint with a constraint name on column sub_category_id of sub_category_details table.

```
ALTER TABLE Sub_category_details
ADD CONSTRAINT pk_sub_category PRIMARY KEY (sub_category_id);
```

3. Add a foreign key constraint with constraint name on column category_id of sub_category_details table referencing category_id of category_details table.

```
ALTER TABLE Sub_category_details
ADD CONSTRAINT fk_sub_category FOREIGN KEY (category_id) REFERENCES
Category_details(category_id);
```

4. For the product_details table add the primary key constraint on product_id. Also add foreign key constraints on category_id and sub_category_id columns referencing category_details(category_id) and sub_category_details (sub_category_id). Give appropriate names for all constraints.

```
ALTER TABLE Product_details
ADD CONSTRAINT pk_product_id PRIMARY KEY (product_id);

ALTER TABLE Product_details
ADD CONSTRAINT fk_category_id FOREIGN KEY (category_id) REFERENCES
category_details(category_id);

ALTER TABLE Product_details
ADD CONSTRAINT fk_sub_category_id FOREIGN KEY (sub_category_id) REFERENCES
sub_category_details(sub_category_id);
```

5. Add a new column (price numeric(2)) to product_details table

```
ALTER TABLE Product_details
ADD COLUMN price NUMERIC(2);
```

6. Modify the data type of price to numeric(6,2)

```
ALTER TABLE Product_details
```

```
MODIFY price NUMERIC(6,2);
```

7. Insert four tuples in the table. (With valid data)

```
INSERT INTO category_details
(category_id, category_name)
VALUES
   (1, 'Electronics'),
   (2, 'Clothing');

INSERT INTO Sub_category_details
(sub_category_id, category_id, sub_category_name)
VALUES
   (14, 1, 'Ear pods'),
   (17, 2, 'T-shirts');

INSERT INTO Product_details
(product_id, category_id, sub_category_id, product_name, price)
VALUES
   (114001, 1, 14, 'One Plus Nord Buds', 3000.00),
   (114002, 1, 14, 'Boat 15', 1000.00),
   (217001, 2, 17, 'DNMX S T-Shirt', 450.00),
   (217002, 2, 17, 'Puma M T-Shirt', 750.00);
```

8. Drop the price column

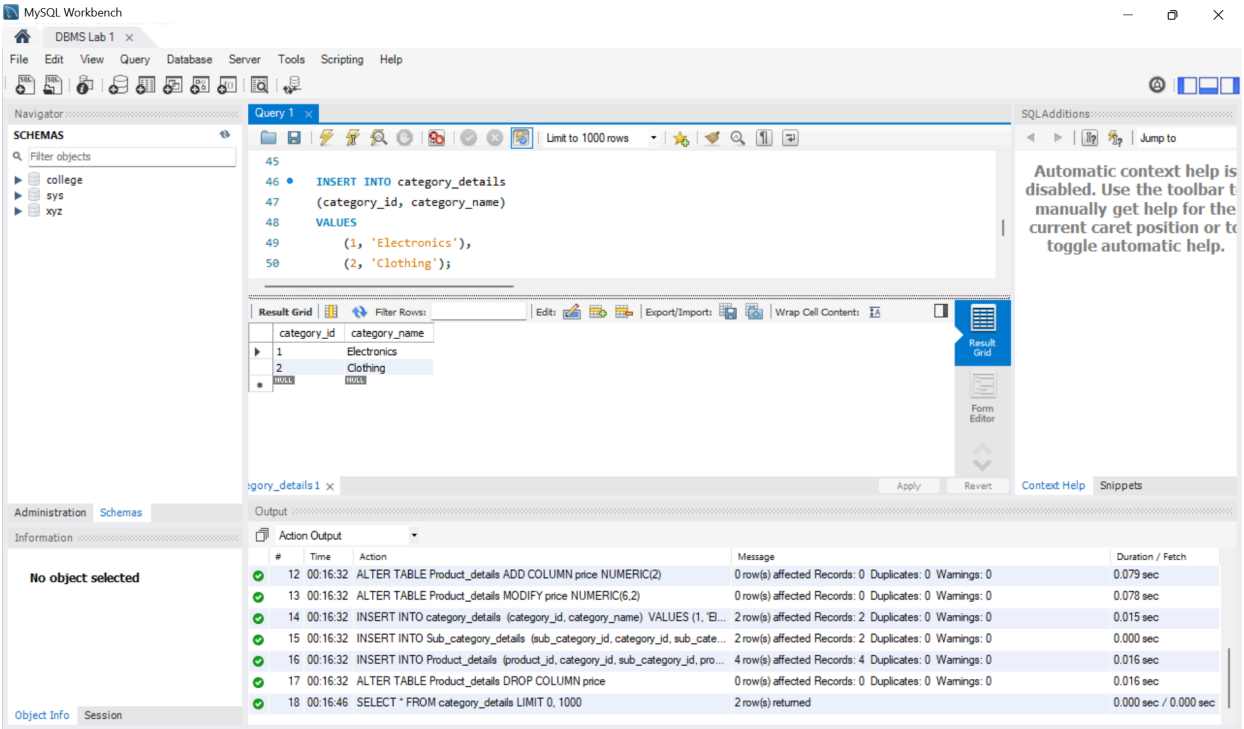
```
ALTER TABLE Product_details
DROP COLUMN price;

# This is to display the tables
SELECT * FROM category_details;
SELECT * FROM sub_category_details;
SELECT * FROM product_details;
```

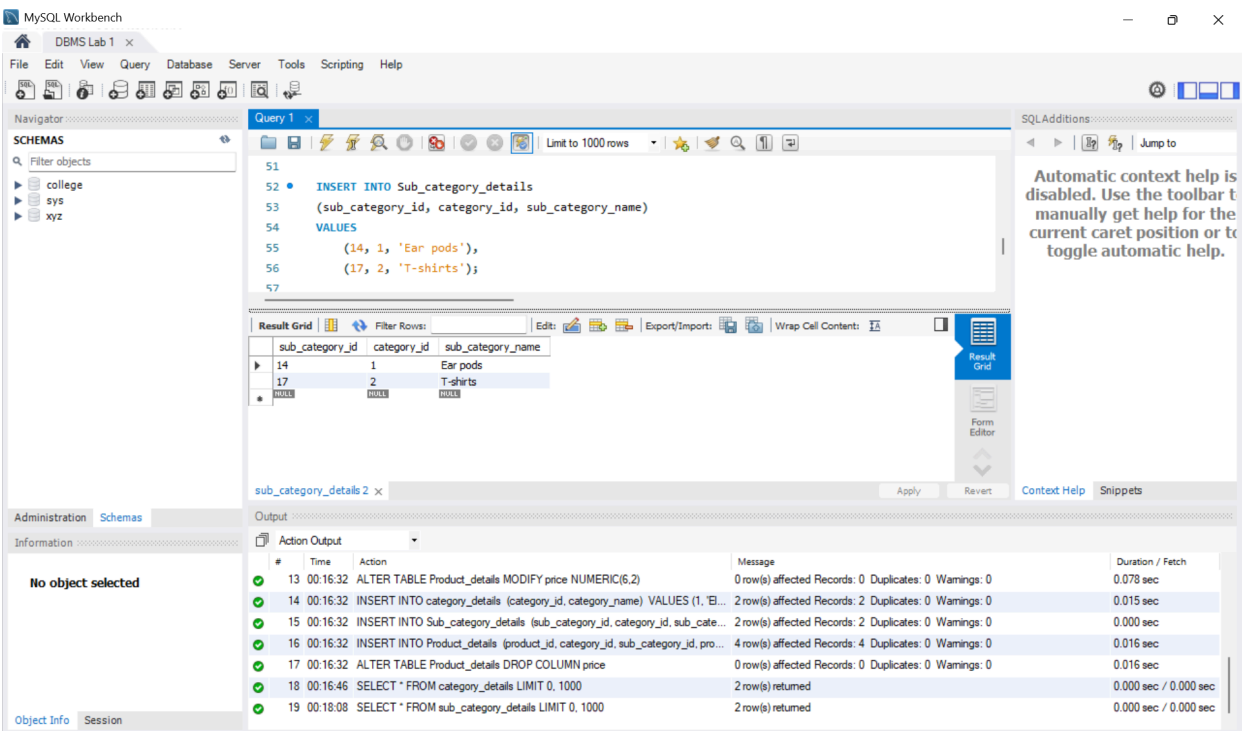
OUTPUTS:

The three output tables are as follows:

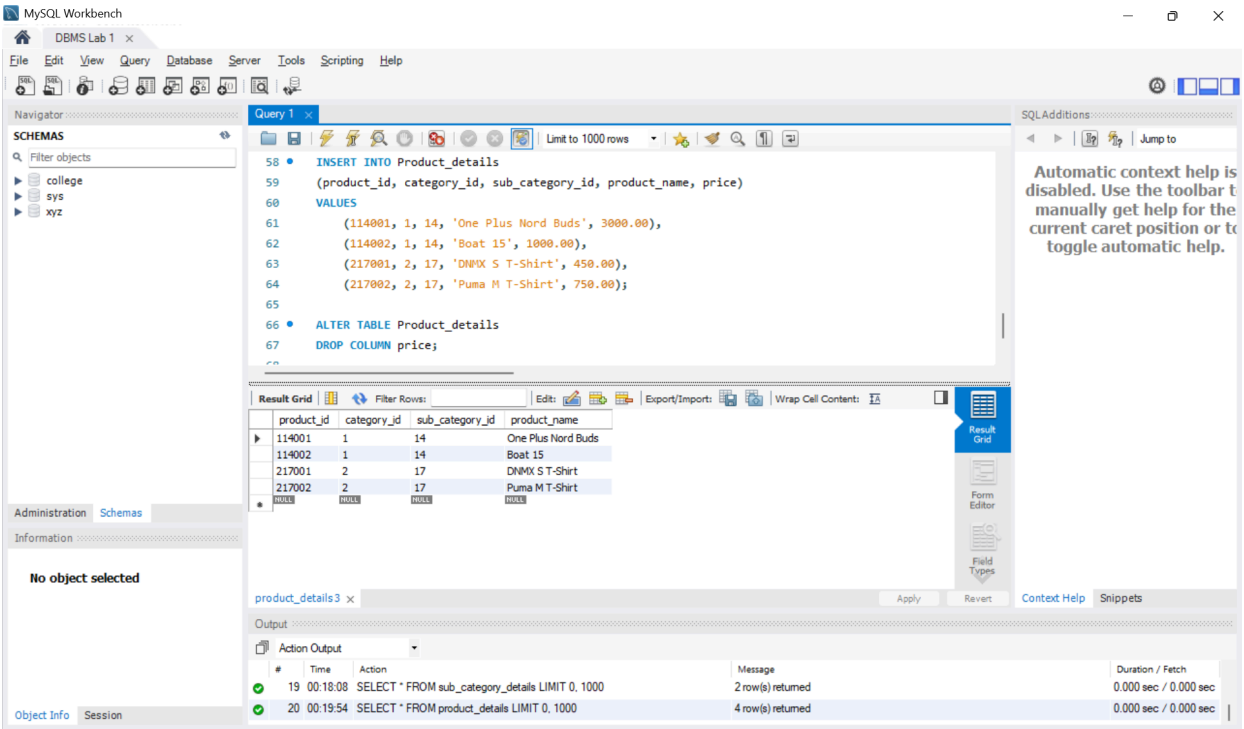
1. Category_details table:



2. Sub_category_details table:



3. Product_details table:



EXECUTION FLOW:

The image illustrates step-by-step execution flow of the above given code

1	15:08:18	CREATE DATABASE product_management_db	1 row(s) affected	0.047 sec
2	15:08:18	USE product_management_db	0 row(s) affected	0.016 sec
3	15:08:18	CREATE TABLE Category_details(category_id NUMERIC(2), category_name ...	0 row(s) affected	0.031 sec
4	15:08:18	CREATE TABLE Sub_category_details(sub_category_id NUMERIC(2), category...	0 row(s) affected	0.031 sec
5	15:08:18	CREATE TABLE Product_details(product_id NUMERIC(6), category_id NUM...	0 row(s) affected	0.032 sec
6	15:08:18	ALTER TABLE Category_details ADD PRIMARY KEY (category_id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
7	15:08:18	ALTER TABLE Sub_category_details ADD CONSTRAINT pk_sub_category PRI...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.046 sec
8	15:08:18	ALTER TABLE Sub_category_details ADD CONSTRAINT fk_sub_category FO...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.094 sec
9	15:08:18	ALTER TABLE Product_details ADD CONSTRAINT pk_product_id PRIMARY K...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
10	15:08:18	ALTER TABLE Product_details ADD CONSTRAINT fk_category_id FOREIGN K...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.093 sec
11	15:08:19	ALTER TABLE Product_details ADD CONSTRAINT fk_sub_category_id FOREI...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.094 sec
12	15:08:19	ALTER TABLE Product_details ADD COLUMN price NUMERIC(2)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
13	15:08:19	ALTER TABLE Product_details MODIFY price NUMERIC(6,2)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.078 sec
14	15:08:19	INSERT INTO category_details (category_id, category_name) VALUES (1, 'Ele...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
15	15:08:19	INSERT INTO Sub_category_details (sub_category_id, category_id, sub_categ...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
16	15:08:19	INSERT INTO Product_details (product_id, category_id, sub_category_id, produ...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	0.016 sec
17	15:08:19	ALTER TABLE Product_details DROP COLUMN price	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
18	15:09:23	SELECT * FROM category_details LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
19	15:10:40	SELECT * FROM sub_category_details LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
20	15:11:29	SELECT * FROM product_details LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
21	15:21:57	SELECT * FROM product_details LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec